International Journal of Forecasting 38 (2022) 321–338

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

A Bayesian approach for predicting food and beverage sales in staff canteens and restaurants☆

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ARTICLE INFO

Keywords:
Sales prediction
Time series
Generalized additive models
Bayesian methods
Shrinkage priors

ABSTRACT

Accurate demand forecasting is one of the key aspects for successfully managing restaurants and staff canteens. In particular, properly predicting future sales of menu items allows for a precise ordering of food stock. From an environmental point of view, this ensures a low level of pre-consumer food waste, while from the managerial point of view, this is critical to the profitability of the restaurant. Hence, we are interested in predicting future values of the daily sold quantities of given menu items. The corresponding time series show multiple strong seasonalities, trend changes, data gaps, and outliers. We propose a forecasting approach that is solely based on the data retrieved from point-of-sale systems and allows for a straightforward human interpretation. Therefore, we propose two generalized additive models for predicting future sales. In an extensive evaluation, we consider two data sets consisting of multiple time series collected at a casual restaurant and a large staff canteen and covering a period of 20 months. We show that the proposed models fit the features of the considered restaurant data. Moreover, we compare the predictive performance of our method against the performance of other well-established forecasting approaches.

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1. Introduction

Total sales in the United States restaurant sector are projected to reach $863 billion in 2019, contributing to 4% of the gross domestic product (National Restaurant Association, 2019). Currently, the restaurant industry employs 15.3 million people at one million locations across the United States. At the same time, the industry accounts for 11.4 million tons of food waste that constitutes an estimated value of $25 billion going to waste every year (ReFED, 2019).

Food waste is a critical socioeconomic problem, considering that 11.8% of households in the United States suffered from food insecurity in 2018 (Coleman-Jensen et al., 2019). Furthermore, the production, transportation, and disposal of unused food significantly impact the environment; see Tonini et al. (2018) and Hickey and Ozbay (2014). On the other hand, food costs represent 28% to 35% of sales in restaurants. Hence, reducing food waste can be critical to boost profitability and to reduce the environmental impact of operating a restaurant. Furthermore, food service contractors face similar challenges when providing meals at staff canteens, hospitals, etc.

Accurate demand forecasting is one of the key aspects for successfully managing restaurants and, from an environmental point of view, maintaining a low level of pre-consumer food waste. In this work, we are interested in predicting future values of the daily sold quantities of a given menu item. Hence, we deal with time series that

☆ The research was by funded by the Austrian Research Promotion Agency (FFG) through the General Programme under the FFG grant number 867779.

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https://doi.org/10.1016/j.ijforecast.2021.06.001
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show multiple strong seasonalities, trend changes, data gaps, and outliers. Traditionally, judgmental forecasting techniques based on the manager's experience are applied to estimate future demand at restaurants. However, producing high-quality forecasts is a time consuming and challenging task, especially for inexperienced managers. Hence, we aim for a data-driven approach that supports restaurant managers in their decision making.

Nowadays, most restaurants and canteens use (electronic) point-of-sale (POS) systems to keep track of all their sales. Clearly, the resulting data inventories are valuable sources for various data science applications, such as forecasting future sales of menu items. In general, many industries rely on POS data to predict future demand. However, especially in the retailing sector, it is hard to make beneficial use of these estimates, due to complicated supply chains and long lead times. In contrast to that, the restaurant industry is characterized by short lead times when ordering food stock and the absence of a complex supply chain.

Hence, we propose a forecasting approach that requires only POS data; i.e., weather information and special promotions are not considered. Moreover, in order to ensure acceptance of the approach among restaurant managers, the models ability for a straightforward human interpretation is critical.

The remainder of this paper is organized as follows. First, in Section 2, we provide a concise problem description. In Section 3, we briefly discuss related literature. Then, in Section 4, we propose our forecasting approach. In Section 5, we evaluate the prediction quality of our approach using real-world data of sales in a restaurant and a canteen. Finally, Section 6 concludes the paper and outlines future research directions.

2. Problem description

This work is concerned with predicting the daily sold quantities of given menu items. This information, together with the recipes of the items, is essential to effectively manage the ordering of food stock. In our use case, a POS system is the exclusive data source. Typically, such systems create a new line item for each sold menu item, together with a time stamp. This way, large data inventories, consisting of thousands of line items, are created. Each restaurant, whether it be a small burger joint or a large company canteen, usually has many different products on its menu. While some products are offered all year long, others are only seasonally available. Moreover, the demand for certain products differs over the year.

Each menu item is usually identified through a unique ID. From the stored records, we can query the accumulated number of sales of the different menu items for each day. Hence, gathering the data required for our forecasting approach causes little to no additional administrative burden. In some cases, it is reasonable to do forecasting for product groups rather than for individual menu items. For example, a canteen may have a daily vegetarian option. Each day, a different dish, identified by its own ID, is served.

2.1. Assumptions

For now, we assume that the sales of individual menu items are independent of each other. While this assumption is unlikely to be true for all types of restaurants, e.g., some sides go along better with certain main dishes than others, it is a reasonable simplifying assumption.

By design, a POS system only documents sales. The absence of the recorded sales of a certain menu item on a given day may be due to several reasons:

(i) The restaurant was closed. In this case, there is no recorded sale of any item at all.
(ii) The item was not on the menu that day by choice or due to the lack of stock.
(iii) No one ordered the product.

Obviously, it makes a difference if a product was not sold because no one ordered it (iii), or because the product was not offered (i) and (ii). To distinguish between these two cases, we treat an absence of sales caused by (iii) as zero sales and the absence of sales caused by (i) or (ii) as missing values. Setting the number of sales independently of (i), (ii), or (iii) as always equal to zero can lead to inferior model fits, especially if the periods where the restaurant is closed or does not offer special products are aperiodic or do not show a common pattern. This could be prevented by adding some information to the model regarding why there are no sales. However, this would require managing and storing additional data other than the POS data, which we do not want to presuppose in this work.

We assume that the restaurant was open for business on a given day if there is at least one recorded sale of any menu item. Usually, data available from a POS system contains no records about when a product was added to and when it was removed from the menu. Therefore, we assume that an item has been removed from the menu if there is no recorded sale for 60 or more days.

3. Related work

In this section, we give a brief overview of related literature. We focus on forecasting approaches used in the restaurant industry, as well as on the use of POS data.

A general view of forecasting methods for restaurant sales and customer demand is given in the recent review paper by Lasek et al. (2016). Ryu and Sanchez (2003) compare several methods (moving average, multiple regression, and exponential smoothing) in order to estimate the daily dinner counts at a university dining center. In their case, they found that multiple regression gives the most accurate predictions. Also, Reynolds et al. (2013) use multiple regression to predict the annual sales volume of the restaurant industry (and of certain subsegments). Forst (1992) applies ARIMA and exponential smoothing models to forecast the weekly sales (in USD) at a small campus restaurant. Hu et al. (2004) use ARIMA methods to predict the customer count at casino buffets. Cranage and Andrew (1992) use exponential smoothing to predict the monthly sales (in USD) at a restaurant. Bujisic et al.

(2017) analyze the effects of weather factors on restaurant sales. Their paper also contains a review of forecasting approaches for restaurant sales (with and without weather factors). However, they point out that most work from the literature is concerned with aggregated forecasts. That is, only predictions for product categories or weekly sales numbers are given, rather than forecasting sales of individual menu items per day.

Moreover, Tanizaki et al. (2019) use machine learning techniques in order to forecast the daily number of customers at restaurants. Their predictions are based on POS data that are enriched with external data, e.g., weather information and event data. Similarly, Kaneko and Yada (2016) present a deep learning approach to construct a prediction model for the sales at supermarkets using POS data.

So far, our short review shows that existing work in the literature is mostly concerned with predicting aggregated numbers, rather than the sale of individual items. Moreover, it reveals that using POS data is common practice in many industries. However, our review exposes a gap in the existing literature. To the best of our knowledge, there is no work concerned with predicting the sales of individual menu items at restaurants (based on POS data).

Hence, we extend our review to prediction approaches that deal with data showing similar features to ours. The problem considered in this work is clearly a time series prediction problem. Its most dominant characteristics are multiple strong seasonalities, trend changes, data gaps, and outliers. Thus, the application of classical time series models, such as ARIMA models or exponential smoothing (see Holt (2004)), is of limited use. These models can only account for one seasonality and, moreover, require missing data points to be interpolated. In principle, multiple seasonality patterns can be dealt with in exponential smoothing; see, e.g., Gould et al. (2008). However, most available ARIMA or exponential smoothing implementations (e.g., Hyndman and Khandakar (2008) and Hyndman et al. (2019)) do not include such features.

Therefore, Taylor and Letham (2018) present a decomposable time series model that is similar to a generalized additive model; see Hastie and Tibshirani (1987). It consists of three main components: trend, seasonality, and holidays. The model is designed in such a way that it allows for a straightforward human interpretation for each parameter, allowing an analyst to adjust the model if necessary. The approach is concerned with time series having features such as multiple strong seasonalities, trend changes, outliers, and holiday effects. As a motivating example, they use a time series that describes the number of events that are created on Facebook every day. However, this number is quite large in comparison to the daily sales in restaurants and canteens. Thus, for the Facebook example, it is valid to assign a normal distribution to the target variable. However, this might be inferior in our use case, since small target values are quite common. Note that the normal distribution $\mathcal{N}(\lambda, \lambda)$ has a shape similar to the Poisson distribution $Poisson(\lambda)$ for large λ. In contrast to the normal distribution, the Poisson distribution is a classical distribution for count data.

4. Methodology

In this section, we propose two Bayesian generalized additive models (GAMs) for demand prediction. The first one assumes a normally distributed target (also called a response) y, and the second one assumes that the response y follows a negative binomial distribution. Both models include a trend function $g(t)$ and a seasonality function $s(t)$. Sparsity-inducing priors (see, e.g., Park and Casella (2008) and Carvalho et al. (2010)) are assigned to the parameters of $g(t)$ and $s(t)$. This allows for good model regularization, selection of significant trend changes or influential seasonal effects, and, finally, enhanced model interpretability.

Considering that restaurants and canteens commonly offer many different menu items and that new data are recorded on a daily basis, multiple models have to be trained on a frequent basis. For this reason, the training process should require little computation time. Additionally, the inference should take place automatically without the drawback that a human analyst has to investigate the convergence of the procedure. Therefore, we consider the application of full Bayesian inference as inappropriate and instead focus on the comparatively easy to obtain mode of the posterior, the so-called maximum a posteriori (MAP) estimate.

4.1. Generalized additive models

A generalized additive model is written as

$$g(\mathbb{E}(y)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + f_1(z_1) + \cdots + f_q(z_q),$$

where y denotes the target variable corresponding to some exponential family distribution, $x_1, \ldots, x_k, z_1, \ldots, x_k \in \mathbb{R}$ denote the predictors (also called covariates), g denotes a link function (bijective and twice differentiable), and $f_1, \ldots, f_q$ denote some smooth functions. Typically, the functions f_i are defined as weighted sums of basis functions, i.e., $f_i = \sum_{k=1}^{K_i} \gamma_{ik} B_{ik}(z_i)$. Sparsity-inducing priors can be used to perform variable selection within the set $\{x_1, \ldots, x_k\}$ and, further, appropriate priors can be used to control the smoothness of the functions $f_1, \ldots, f_q$.

4.2. Trend function

To model the trend of a time series, i.e., a series of data points ordered in time, we use a polynomial spline g of degree l; see Fahrmeir et al. (2013).

A mapping $g : [a, b] \to \mathbb{R}$ is called a polynomial spline of degree (order) $l \geq 0$ with knot points at $k_1 < \cdots < k_m$, $(a < k_1, k_m < b)$, if

1. g is a polynomial of degree l on each of the intervals $[a, k_1)$, $[k_1, k_2)$,…, $[k_m, b)$, and
2. g is $l-1$ times continuously differentiable, provided that $l > 0$.

Let $S_l^{\mathbf{k}}$ denote the set of all lth order splines with knots given by $k_1 < \cdots < k_m$. Equipped with the operations of adding two functions and taking real multiples, $S_l^{\mathbf{k}}$ is a real vector space. One can show that each element of

$S_l^{\mathbf{k}}$ can be uniquely written as linear combination of the $d = l + m + 1$ functions

$$B_1(t) = 1,\ B_2(t) = t, \ldots,\ B_{l+1}(t) = t^l,$$
$$B_{(l+2)}(t) = (t - k_1)_+^l, \ldots,\ B_{(l+m+1)}(t) = (t - k_m)_+^l,$$

where

$$(t - k_j)_+^l = \begin{cases} (t - k_j)^l, & t \geq k_j, \\ 0, & \text{else.} \end{cases}$$

For this reason, the functions $B_1, \ldots, B_d$ build a basis B of the spline space $S_l^{\mathbf{k}}$. B is called the truncated power series basis (TP-basis). The TP-basis allows for a simple interpretation of the trend model

$$\begin{aligned} g(t) &= \sum_{j=1}^{d} \gamma_j B_j(t) \qquad (1) \\ &= \gamma_1 + \gamma_2 t + \cdots + \gamma_{l+1} t^l + \gamma_{l+2}(t - k_1)_+^l \\ &\quad + \cdots + \gamma_{l+m+1}(t - k_m)_+^l. \end{aligned}$$

The trend model g consists of a global polynomial of degree l which changes at each knot k_j. The amount of change at a given knot point k_j is determined by the absolute value of the corresponding coefficient γ_{l+j+1}. Thus, the knots of the polynomial spline are interpreted as (possible) change points in the trend.

As a consequence, the assignment of a sparsity-inducing prior to the trend coefficients $\gamma_1, \gamma_2, \ldots, \gamma_d$ would allow for an identification of truly significant trend changes. This increases the interpretability of the model and regularizes the trend function.

According to domain experts, for most menu items, the trend (regarding the number of sales) is quite constant over long periods, while significant changes appear only occasionally. Hence, we decide to specify a knot point every kth day between the first and the last date with observations. Assuming that the trend may change on a weekly basis, a possible value for k could be $k = 7$. However, for restaurants that observe only slight changes once in a while, larger values for k are also useful in order to reduce the size of the resulting model. Consequently, we choose $k = 30$ in our experimental evaluation; see Section 5.4.3.

4.2.1. Prior for the trend

In this section, we propose two different priors for the trend model. The first one is mainly inspired by the Bayesian Lasso (Park & Casella, 2008). Here, we assign independent Laplace priors to the coefficients of the trend model that do not belong to the global polynomial. In case y is assumed to be normally distributed with given variance σ^2, the priors read as

$$\gamma_{l+2}|\sigma^2, \ldots, \gamma_d|\sigma^2 \underset{i.i.d.}{\sim} Laplace\left(0, \frac{\sqrt{\sigma^2}}{\tau_1}\right). \qquad (2)$$

The conditioning on σ^2 is important, because it guarantees a unimodal full posterior; see Park and Casella (2008). Similarly, in the case of a negative binomial response y, the priors are given by

$$\gamma_{l+2}, \ldots, \gamma_d \underset{i.i.d.}{\sim} Laplace\left(0, \frac{1}{\tau_1}\right). \qquad (3)$$

Assigning Laplace priors to coefficients $\gamma_{l+2}, \ldots, \gamma_d$ results in a sparse MAP estimate of these parameters. The amount of sparsity is controlled by the tuning parameter τ_1. As a consequence, true change points can be automatically detected, while wrongly proposed ones are ignored. Moreover, as in Bayesian ridge regression, we assign independent normal priors to the coefficients $\gamma_2, \ldots, \gamma_{l+1}$:

$$\gamma_2, \ldots, \gamma_{l+1} \underset{i.i.d.}{\sim} \mathcal{N}(0, \tau_3^2). \qquad (4)$$

Thus, in the MAP estimate, each of these coefficients is shrunken towards zero according to its importance in a matter of regularization. Finally, the improper and uninformative prior $p(\gamma_1) \propto 1$ is assigned to the intercept of the trend function.

Besides the usage of Laplace and normal priors, we also propose assigning the horseshoe prior (see Carvalho et al. (2010) to all the coefficients of the trend function except that of the intercept. In the case of a Gaussian target variable, the prior reads as follows (see Makalic and Schmidt (2016):

$$\begin{aligned} \gamma_j|\lambda_j, \tau, \sigma^2 &\sim \mathcal{N}(0, \lambda_j^2 \tau^2 \sigma^2), \\ \lambda_j &\sim C^+(0, 1), \\ \tau &\sim C^+(0, 1), \end{aligned}$$

where C^+ denotes the half-Cauchy distribution and $j \in \{2, \ldots, d\}$. If a negative binomial response is assumed, σ^2 has to be removed from the above specification. The horseshoe prior is a shrinkage prior which enforces a global scale τ on the one hand and, on the other hand, allows for individual adaptions $\lambda_2, \ldots, \lambda_d$ of the degree of overall shrinkage.

The possibility of using individual shrinkage strengths is sometimes an advantage of the horseshoe prior over the Bayesian Lasso, where the shrinkage effect is uniform across all coefficients. We note that for some menu items the trend of the corresponding time series changes drastically within a short period of time. In this case, uniform shrinkage leads either to an over-shrinkage of heavy changes or to an under-shrinkage of insignificant changes. However, it turned out that a model using the horseshoe prior requires a very good initialization of the respective optimization routine, in order for the procedure to converge towards a useful MAP estimate. Since we could not identify an automatic approach to find such an initialization, we recommend using the Bayesian Lasso prior as a standard approach, and letting an expert apply the horseshoe prior if necessary.

Taking into account that the application of the horseshoe prior leads to problems in terms of automation, we did not consider other similarly sophisticated priors, e.g., (Bhattacharya et al., 2015; Rockova & George, 2018). However, these will be considered in future work.

4.2.2. Dates

Obviously, the trend model requires the time t to be given as numeric values. Hence, we use a function $\delta : \{dates\} \rightarrow \mathbb{R}$ that maps all possible dates (time points, days) to real numbers. We assume that observations are available at the dates $t_1 < \cdots < t_n$. Further, let $N \geq n$ denote the number of days that lie in the interval $[t_1, t_n]$.

Then, δ maps t_1 to zero and each other date t to the number of days that t_1 differs from t divided by N. Finally, $\delta(t_1), \ldots, \delta(t_n)$ are elements of the interval $[0, 1)$.

4.3. Seasonality function

The seasonality function $s(t)$ is used to model periodic changes of the daily sold quantities y_t. In order to take into account seasonal effects—the day of the week, the day of the month, and the month of the year—diverse dummy variables are introduced (taking values of only 0 and 1 in order indicate the presence of a given season). Let $\mathbf{x} = (x_1, \ldots, x_s)^\mathsf{T}$ denote the vector consisting of all dummy variables corresponding to the considered seasonal effects. The seasonality function is then given by

$$s(t) = \beta_1 x_1(t) + \cdots + \beta_s x_s(t) = \boldsymbol{\beta}^\mathsf{T}\mathbf{x}(t).$$

4.3.1. Prior for the seasonality

Inspired by the Bayesian Lasso (Park & Casella, 2008), we assign independent Laplace priors to the coefficients of the seasonality model. For a normally distributed target y, the priors are specified as

$$\beta_1|\sigma^2, \ldots, \beta_s|\sigma^2 \underset{i.i.d.}{\sim} \mathit{Laplace}\left(0, \frac{\sqrt{\sigma^2}}{\tau_2}\right),$$

and for a negative binomial target, the priors read as

$$\beta_1, \ldots, \beta_s \underset{i.i.d.}{\sim} \mathit{Laplace}\left(0, \frac{1}{\tau_2}\right).$$

The sparsity of the parameter's MAP estimate induced by the Laplace priors enables the automatic selection of influential seasonal effects. We do not consider the usage of the horseshoe prior (see Carvalho et al. (2010) for the seasonal model. As mentioned in Section 4.2.1, this prior results in a more complicated model optimization. Since we empirically discovered that the Bayesian Lasso prior performs quite well for the seasonal model, it appears unnecessary to accept the additional complexity.

4.3.2. Deciding the granularity of the seasonality model

Let the number of days with observations be denoted by n. Clearly, the size of the number n influences the usefulness of modeling diverse seasonal effects. Hence, we apply the following strategy:

- $n < 30$: model only the effect of the day of the week,
- $30 \leq n < 120$: additionally model the effect of the month of the year,
- $n >= 120$: model all seasonal effects (day of the week, day of the month, and month of the year).

4.4. Prediction models

As mentioned at the beginning of Section 4, we propose two Bayesian GAMs for predicting the daily sold quantities of certain menu items in restaurants and canteens. Both models include a trend function $g(\delta(t))$ (see Section 4.2) and a seasonality function $s(t)$ (see Section 4.3).

4.4.1. Normal model

In the so-called normal model, the daily sold quantity at time t, denoted by y_t, is modeled as

$$y_t = g(\delta(t)) + s(t) + \varepsilon_t, \tag{5}$$

where ε_t denotes noise terms that are assumed to be i.i.d. according to $\mathcal{N}(0, \sigma^2)$. Additionally, the non-informative and scale-invariant prior $p(\sigma^2) \propto \frac{1}{\sigma^2}$ is assigned to the unknown error variance. Suggestions for priors used to regularize $g(\delta(t))$ and $s(t)$ are described in Sections 4.2.1 and 4.3.1. Assuming that we have observations at time points $t_1 < \cdots < t_n$, then Eq. (5) translates to

$$\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\gamma}, \sigma^2 \sim \mathcal{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}, \sigma^2\mathbf{I}), \tag{6}$$

with

$$\mathbf{X} = \begin{pmatrix} x_1(t_1) & x_2(t_1) & \cdots & x_s(t_1) \\ \vdots & \vdots & & \vdots \\ x_1(t_n) & x_2(t_n) & \cdots & x_s(t_n) \end{pmatrix}, \tag{7}$$

according to Section 4.3, and

$$\mathbf{Z} = \begin{pmatrix} 1 & \delta(t_1) & \cdots & \delta(t_1)^l & (\delta(t_1) - \delta(k_1))_+^l & \cdots (\delta(t_1) - \delta(k_m))_+^l \\ \vdots & \vdots & & \vdots & \vdots & \vdots \\ 1 & \delta(t_n) & \cdots & \delta(t_n)^l & (\delta(t_n) - \delta(k_1))_+^l & \cdots (\delta(t_n) - \delta(k_m))_+^l \end{pmatrix}, \tag{8}$$

according to Section 4.2.

4.4.2. Negative binomial model

As can be seen in Eq. (6), the assumption of additive and normally distributed noise directly implies that the target y also follows a normal distribution. In our use case, this assumption cannot be satisfied, since the target is a non-negative integer. Indeed, this can even lead to predicting negative values, in case the trend is a monotonically decreasing function. More appropriate distributions for the target variable are the Poisson distribution and the negative binomial distribution, which are both well-established distributions for regression with count data.

Nevertheless, it should be mentioned that, for large λ, the normal distribution $\mathcal{N}(\lambda, \lambda)$ is a good approximation of the Poisson distribution $\mathit{Poisson}(\lambda)$, due to the central limit theorem. However, the mean number of daily sold quantities can be quite small for some menu items in restaurants and staff canteens. Thus, we model the target y with a normal distribution and with a negative binomial distribution. This distribution can be considered as an extension of the Poisson distribution that also allows for over-dispersion. The mean and the variance of y are given by

$$\mathbb{E}(y) = \mu,$$
$$\mathrm{Var}(y) = \mu + \mu^2/\phi,$$

where $\mu, \phi \in \mathbb{R}^+$ denote the parameters of the distribution.

Although, the negative binomial distribution does not allow for under-dispersion, we do not consider this limitation as a major drawback. Under-dispersed data are

rather unlikely in practical applications; see Hilbe et al. (2017). Domain experts have confirmed this statement with regards to the application considered in this work. In other applications, where under-dispersion is common, the Conway–Maxwell–Poisson distribution (see Conway and Maxwell (1962)) is a notable alternative to the negative binomial distribution. This distribution allows for both under-dispersed and over-dispersed data, but at the price of significantly increased model complexity.

In the so-called negative binomial model, the quantity y_t sold on day t is modeled as

$$y_t|\boldsymbol{\beta}, \boldsymbol{\gamma}, a \sim NegBinom\left(\mu_t = \exp(g(\delta(t)) + s(t)), \phi = \frac{1}{a^2}\right), \quad (9)$$

with $a > 0$. Using the exponential function exp() as the response function ensures that the expected demand μ_t always stays positive, regardless of the estimates of $g(\delta(t))$ and $s(t)$. Another side effect of this response function is that the seasonal component acts in a multiplicative way on the trend, $\mu_t = \exp(g(\delta(t)))\exp(s(t))$. As recommended by Gelman (2019), we assign a standard half-normal prior $\mathcal{N}^+(0, 1)$ to $a = \frac{1}{\sqrt{\phi}}$:

$$a \sim \mathcal{N}^+(0, 1). \quad (10)$$

Note that by assigning the prior $\mathcal{N}^+(0, 1)$ directly to ϕ, most of the prior mass would be on models with a large amount of over-dispersion. In the case of limited over-dispersion, this can lead to a conflict between prior and data.

Assuming that observations are available at the time points $t_1 < \cdots < t_n$, model (9) translates to

$$\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\gamma}, a \sim \prod_{i=1}^{n} \text{NegBinom}\left(\exp\left\{\mathbf{X}[i,:]\boldsymbol{\beta} + \mathbf{Z}[i,:]\boldsymbol{\gamma}\right\}, \frac{1}{a^2}\right), \quad (11)$$

where $\mathbf{y} = (y_{t_1}, \ldots, y_{t_n})^\top$, $\mathbf{X}[i,:]$ denotes the ith row of matrix $\mathbf{X}$ defined via (7), and $\mathbf{Z}[i,:]$ denotes the ith row of matrix $\mathbf{Z}$ defined via (8).

4.5. Choice of tuning parameters

The priors proposed for the seasonality function $s(t)$ (see Section 4.3.1) depend on a hyperparameter τ_2. Further, in the case that the priors (2)–(4) are used for the trend $g(\delta(t))$, they depend on hyperparameters denoted by τ_1 and τ_3. These parameters can be considered as tuning parameters that control the amount of sparsity/regularization in the estimates of $g(\delta(t))$ and $s(t)$. Thus, a good specification of τ_1, τ_2, and τ_3 is essential for model performance. Bad choices could allow for too much flexibility within a given model (inclusion of too many seasonal effects, a trend function that nearly interpolates the data, etc.) and, thus, result in overfitting. Or they could allow for too little flexibility, such that some significant effects are ignored.

In principle, there are three approaches to specifying the hyperparameters τ_1, τ_2, τ_3:

- The first approach is to determine standard settings that perform well on average.
- Another possibility is to determine useful settings via cross-validation. While, in general, the prediction quality of the resulting model is better than for the standard settings, the process itself is computationally expensive.
- The third option is to specify hyper-priors for the tuning parameters.

In this work, we focus on the first two approaches, and defer the last one to future work.

4.5.1. Standard settings

Certainly, good specifications of the tuning parameters differ from data set to data set—i.e., in the context of this work, from menu item to menu item. Nevertheless, it is useful to have a fixed specification of them that works well on average. For this purpose, we tested diverse possible specifications of these parameters with several data sets of representative menu items. The following specifications led to the best results:

$$\tau_1 = 5, \qquad \tau_2 = 6, \qquad \tau_3 = \begin{cases} 0.001, & \text{if } n < 120, \\ 0.01, & \text{if } 120 \leq n < 350, \\ 0.5, & \text{else.} \end{cases} \quad (12)$$

4.5.2. Cross-validation

Cross-validation allows for an automatic identification of good and individual tuning parameter specifications for each data set. Applying cross-validation on a given data set means that the data set is partitioned into multiple training and testing splits. On each of these splits, the considered model is trained and tested with different specifications of its tuning parameters. Finally, we select the specification which on average corresponds to the best predictions (according to some pre-defined quality measure). To account for temporal dependencies, we consider an expanding window approach to define the training and testing splits. The approach is outlined in Fig. 1. Looking at the figure, the amount of training data is gradually reduced, while the amount of testing data is kept constant.

4.5.3. Overall strategy

We recommend using the standard settings proposed in Section 4.5.1, and performing cross-validation only for the most important menu items when model performance is insufficient. The restriction to the standard settings should not be considered too critical, since we discovered empirically that they perform quite well.

In the case that cross-validation is applied to specify τ_1, τ_2, τ_3, we perform the process step by step for each single tuning parameter. Optimizing all three tuning parameters at once would require training substantially more models and is therefore considered too computationally expensive. Within the step-wise procedure, the parameter τ_3 (responsible for the global trend) is first optimized, followed by the parameter τ_1 (responsible for

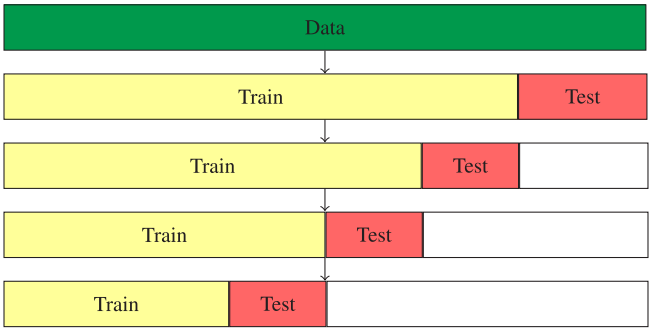


Fig. 1. Illustration of the cross-validation approach. The data are assumed to be temporally ordered.

trend changes), and, finally, the parameter τ_2 (responsible for seasonal effects). Parameters that are not currently optimized are either fixed with the standard settings provided in (12), or, if available, with the already optimized values.

Finally, it should be mentioned that the quality criterion used inside the cross-validation is the mean absolute deviation (MAD) between the predicted values and the corresponding true ones.

4.6. Prediction & prediction uncertainty

Assume that y_t has to be predicted for $t \in \{t_1^*, \dots, t_p^*\}$ with $t_1^* < \cdots < t_p^*$ and, further, assume that observations are available at the time points $t_1 < \cdots < t_n$ with $t_1 \leq t_1^*$. Let $\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}}, \widehat{\sigma}$, and $\widehat{a}$ denote MAP estimates corresponding either to the normal model (6) or to the negative binomial model (11). Additionally, treat the estimates as if they were the true values. Then the expected value

$$\begin{aligned}\boldsymbol{\mu}^* &= \mathbb{E}(\mathbf{y}^* = (y_{t_1^*}, \dots, y_{t_p^*})^\mathsf{T} | \widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}}),\\ &= \begin{cases} \mathbf{X}^*\widehat{\boldsymbol{\beta}} + \mathbf{Z}^*\widehat{\boldsymbol{\gamma}}, & \text{normal model},\\ \exp(\mathbf{X}^*\widehat{\boldsymbol{\beta}} + \mathbf{Z}^*\widehat{\boldsymbol{\gamma}}), & \text{negative binomial model}, \end{cases}\end{aligned} \tag{13}$$

can be used for prediction. Note that the exponential function in Eq. (13) is taken component-wise and, moreover, that $\mathbf{X}^*$ and $\mathbf{Z}^*$ are defined analogously to $\mathbf{X}$ and $\mathbf{Z}$ (Eqs. (7) and (8)), by replacing $t_1, \dots, t_n$ with $t_1^*, \dots, t_p^*$.

Besides predicting a posteriori reasonable values for $\mathbf{y}^*$, providing some uncertainty information regarding the forecasts is essential for planning food stock orderings. Intervals that contain the $y_{t_i^*}$ $(i = 1, \dots, p)$ with a predefined probability are of particular interest. Then, depending on the importance of the availability of a given menu item, the manager can decide to go along with the prediction $\mu_i^* = \mathbb{E}(y_{t_i^*} | \widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$, or adjust the orders closer to the upper or lower interval boundary. Recall (Sections 4.2 and 4.2.1) that the trend is assumed to be constant most of the time, with occasional changes. To model this assumption, the trend function $g(\delta(t))$ includes a knot every kth day between the first and the last date with observations. Additionally, sparsity-inducing priors are assigned to the coefficients $\gamma_{l+2}, \dots, \gamma_d$ that represent trend changes at the knot points $k_1, \dots, k_m$. The restriction of merely specifying knots in the observed time period implies that the trend stays constant in unobserved periods, i.e., in the future. However, in case prediction uncertainty information is required, possible future trend changes must be considered. Hence, we extend the trend function to

$$\begin{aligned} g^*(\delta(t)) = \; & \gamma_1 + \gamma_2\delta(t) + \cdots + \gamma_{l+1}\delta(t)^l \\ & + \gamma_{l+2}(\delta(t) - \delta(k_1))_+^l + \\ & \dots + \gamma_d(\delta(t) - \delta(k_m))_+^l + \gamma_{d+1}(\delta(t) \\ & - \delta(k_{m+1}))_+^l + \\ & \dots + \gamma_{d+r}(\delta(t) - \delta(k_{m+r}))_+^l, \end{aligned} \tag{14}$$

where $k_{m+1}, \dots, k_{m+r}$ denote additionally introduced knot points. The additional knots extend the approach of specifying a knot point every kth day within the observed period to the time interval $[t_1, t_p^*] \cup [t_1, t_n]$. When $t_p^* - k_m < k$, the expansion g^* is equal to the original trend function g. Additionally, it is assumed that the coefficients $\gamma_{d+1}, \dots, \gamma_{d+r}$ are i.i.d. zero mean Laplace distributed:

$$\gamma_{d+1}, \dots, \gamma_{d+r} \underset{\text{i.i.d.}}{\sim} \mathit{Laplace}(0, b). \tag{15}$$

Assuming that the scale of future trend changes is determined by the scale of the historically observed changes $\widehat{\gamma}_{l+2}, \dots, \widehat{\gamma}_d$, the parameter b is estimated as

$$\widehat{b} = \frac{1}{m}\sum_{i=1}^{m} |\widehat{\gamma}_{l+i+1}|. \tag{16}$$

Note that given i.i.d. samples $x_1, \dots, x_N$ from a zero mean Laplace distribution with scale parameter b, the mean of the absolute values of the x_i is the maximum likelihood estimator of b. Since the estimates $\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}}, \widehat{\sigma}$, and $\widehat{a}$ are treated as if they were the true values, $(1-\alpha)$ prediction intervals for the $y_{t_i^*}$ $(i = 1, \dots, p)$ can be computed as follows:

1. Expand the trend function to g^*; see Eq. (14).
2. By replacing $t_1, \dots, t_n$ with $t_1^*, \dots, t_p^*$:
   (a) Determine the model matrix $\mathbf{X}^*$ analogous to the computation of $\mathbf{X}$ (Eq. (7)).
   (b) Determine the model matrix $\mathbf{Z}^{**}$ analogous to the computation of $\mathbf{Z}$. Note that that now g^* is used to model the trend and not g.

3. If g^* differs from g, sample from $\boldsymbol{\gamma}^* = (\gamma_{d+1}, \ldots, \gamma_{d+r})^\mathsf{T}$ according to Eqs. (15) and (16). Let the samples be denoted by $\tilde{\boldsymbol{\gamma}}^* = (\tilde{\gamma}_{d+1}, \ldots, \tilde{\gamma}_{d+r})^\mathsf{T}$.
4. Draw a sample from the conditional distribution of $\mathbf{y}^* = (y_{t_1^*}, \ldots, y_{t_p^*})^\mathsf{T}$ conditioned on $\tilde{\boldsymbol{\gamma}}^*$. The distribution is given by:

$$\mathbf{y}^* | \tilde{\boldsymbol{\gamma}}^* \sim \begin{cases} \mathcal{N}(\mathbf{X}^* \widehat{\boldsymbol{\beta}} + \mathbf{Z}^{**}(\widehat{\boldsymbol{\gamma}}^\mathsf{T}, \tilde{\boldsymbol{\gamma}}^{*\mathsf{T}})^\mathsf{T}, \widehat{\sigma}^2 \mathbf{I}), & \text{normal,} \\ \prod_{i=1}^{p} \text{NegBinom}\left(\exp\left\{ \mathbf{X}^*[i,:]\widehat{\boldsymbol{\beta}} + \mathbf{Z}^{**}[i,:](\widehat{\boldsymbol{\gamma}}^\mathsf{T}, \tilde{\boldsymbol{\gamma}}^{*\mathsf{T}})^\mathsf{T} \right\}, \frac{1}{\widehat{a}^2} \right), & \text{neg. bin.} \end{cases}$$

   When g^* equals g, the vector $\boldsymbol{\gamma}^*$ is empty and can be ignored.
5. Repeat step 3 and step 4 a pre-defined number of times.
6. Compute the $\alpha/2$ and $(1-\alpha/2)$ quantiles of the samples drawn from $\mathbf{y}^* | \tilde{\boldsymbol{\gamma}}^*$ component-wise in order to obtain the desired prediction intervals.

It should be mentioned that the assumptions taken for the uncertainty estimation are rather strong and, therefore, one cannot expect the prediction intervals to have exact coverage. Therefore, the intervals should be considered as an indicator of the level of uncertainty.

At first, the restriction to MAP estimates implies that uncertainty is underestimated. However, as stated at the beginning of Section 4, we consider the application of full Bayesian inference to be inappropriate, due to the additional complexity. Moreover, it is restrictive to assume that future trend changes are independent and of the same average magnitude as historical changes. Nevertheless, considering that restaurants and canteens commonly plan their stock orders 7–14 days in advance, the number of possible future trend changes that have to be taken account of is very limited. For this reason, using overly sophisticated approaches to model future trend changes is not required.

In addition to using α% prediction intervals to measure the uncertainty of future predictions, they can also be used to determine whether the model fits the observed data. To do so, the intervals are first computed for the time period with observations. Then, we check whether the fraction of days for which the target variable y does not lie within the corresponding intervals is significantly larger than $(1-\alpha)$%. If this is the case, the model does not fit the data well and must be adjusted.

5. Performance evaluation

In this section, the prediction quality of our approach is evaluated. First, Section 5.1 provides a brief description of the test data used. Then, in Section 5.2, we briefly explain the implementation of the proposed models. In Section 5.3, we evaluate how the models fit the features of the considered restaurant data. Finally, in Section 5.3.2, we compare our prediction models against other promising approaches from the literature.

Throughout this section, the degree l of the polynomial spline used to model the trend is specified as $l = 1$. It turned out that higher degrees often caused overly pronounced trend decreases or increases during prediction. Moreover, the trend function is designed to have a knot point every 30th day.

5.1. Testing data

In this study, POS data from two different restaurants were used. The data were provided by a partnering restaurant consulting firm. Restaurant A is a casual dining establishment located in Vienna, Austria that offers a traditional Austrian menu. Restaurant B is a large staff canteen in the Netherlands operated by a major food services company. For both locations, the data cover a time span of about 20 months. Moreover, the most representative menu items were selected, i.e., those accounting for the most sales. For restaurant A, the menu items were categorized. An overview of the time series data considered per category is given in Table 1. Additionally, this table provides some descriptive statistics of the time series data. In particular, the mean, standard deviation (sd), minimum (min), maximum (max), and the number of missing values (miss) are presented. These statistics were computed for the time series with the highest sales revenue (indicated by H) and the time series with the lowest sales revenue (indicated by L) in each category. The temporal order of the data was ignored when calculating the mean and the standard deviation. The idea of the statistics is simply to provide the reader with an example of the range of values for the individual series. For restaurant B a categorization was not possible, since we only obtained data for which the product names and categories were masked. Similar to restaurant A, we provide an overview of the corresponding time series data in Table 2. Notice that restaurant B shows a high number of missing values, since it is closed on weekends (compare to Section 2.1).

In order to provide some structural characteristics of the POS data, four representative time series (A1, A2, B1, and B2) are considered. A1 and A2 correspond to restaurant A, while B1 and B2 belong to restaurant B. In Figs. 2–5, the time series are decomposed into trends and weekly, monthly, and yearly seasonal patterns. The decomposition is obtained from the proposed model that assumes a negative binomial target; see Section 4.4.2. This model was chosen for the decomposition, since it achieved the best results in the performance evaluation presented in Section 5.4.3. Recall that in this model, the seasonal effects (for weekday, day of the month, and month of the year) act multiplicatively on the trend. Note that the seasonal effects are obtained by passing the coefficients of the seasonality function to the exponential function. Figs. 2–3 show that the trend is often quite constant (A1, A2), while at the same time smooth (B1) and drastic (B2) trend changes appear. Moreover, weekly, monthly, and yearly seasonal patterns are observed. For instance, time series A1 shows an increasing number of sales at the end of the year, time series A2 shows a similar behavior at the end of the month, and time series B1 and B2 show the lowest number of sales on Fridays. Finally, the presented time series also show some outliers.

Table 1
Overview of data from restaurant A per category.

Category	# time series	meanH	meanL	sdH	sdL	minH	minL	maxH	maxL	missH	missL
Starters	5	10.008	3.265	6.927	3.079	0	0	47	17	25	25
Side dishes	7	9.717	1.786	4.865	1.627	0	0	29	9	25	25
Main dishes	21	14.980	1.868	7.294	1.671	0	0	50	9	25	25
Desserts	7	7.151	0.875	5.980	1.373	0	0	41	18	25	25
Snacks	16	20.731	1.235	11.223	1.236	0	0	74	6	25	25
Alcoholic beverages	27	110.706	2.483	45.196	2.864	0	0	350	30	25	12
Non-alcoholic beverages	32	10.627	1.313	6.113	1.469	0	0	48	9	25	12

Table 2
Overview of data from restaurant B.

# time series	meanH	meanL	sdH	sdL	minH	minL	maxH	maxL	missH	missL
30	651.973	19.587	765.046	38.178	0	0	9769	184	231	231

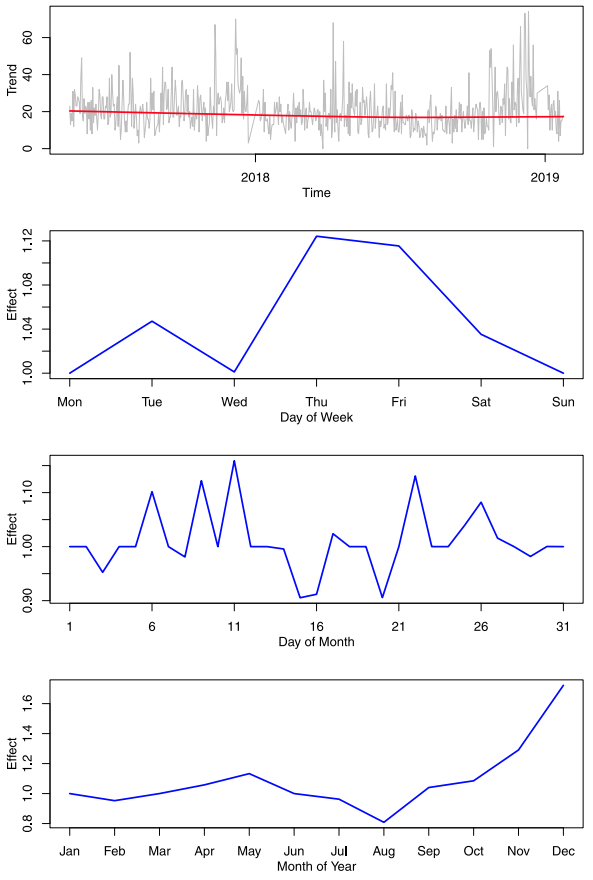


Fig. 2. Structural characteristics of time series A1. The time series is decomposed into trends and weekly, monthly, and yearly seasonal patterns.

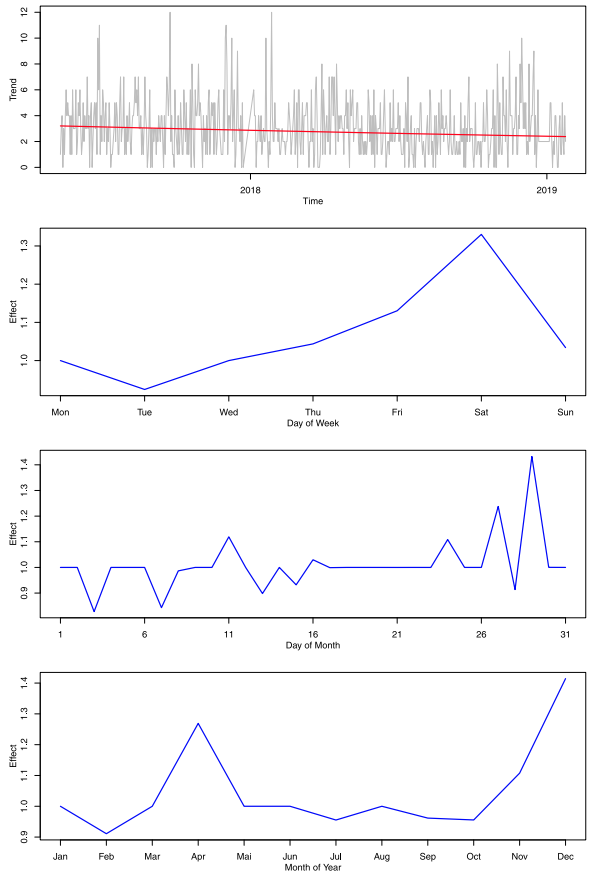


Fig. 3. Structural characteristics of time series A2. The time series is decomposed into trends and weekly, monthly, and yearly seasonal patterns.

It should be mentioned that the outliers observed in these time series are not as strong as outliers we could observe in other time series from restaurant A or restaurant B. For reasons of brevity, we do not include further time series plots in this section.

5.2. Implementation of the models

We implemented the models proposed in Section 4 using *RStan* (Stan Development Team, 2018), the interface between the programming languages *R* (R Core Team,

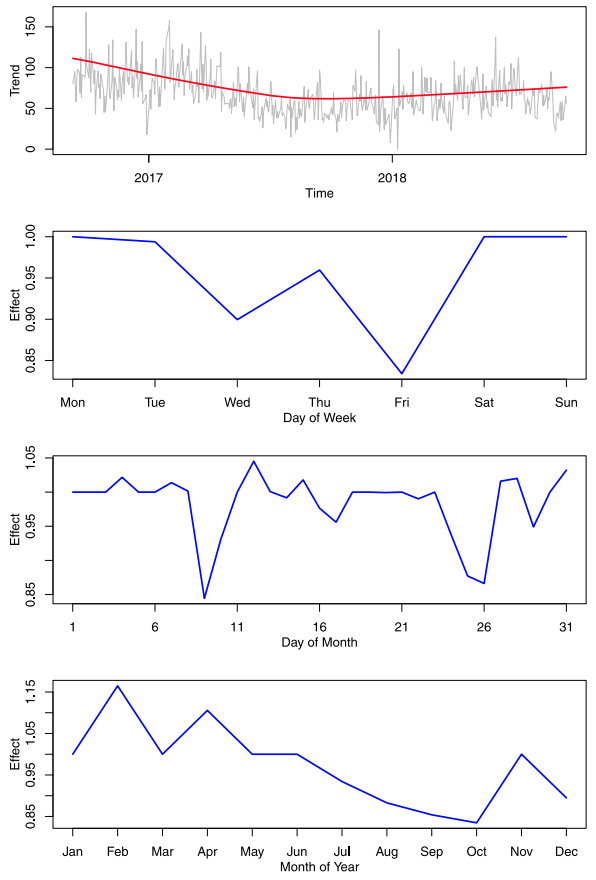


Fig. 4. Structural characteristics of time series B1. The time series is decomposed into trends and weekly, monthly, and yearly seasonal patterns.

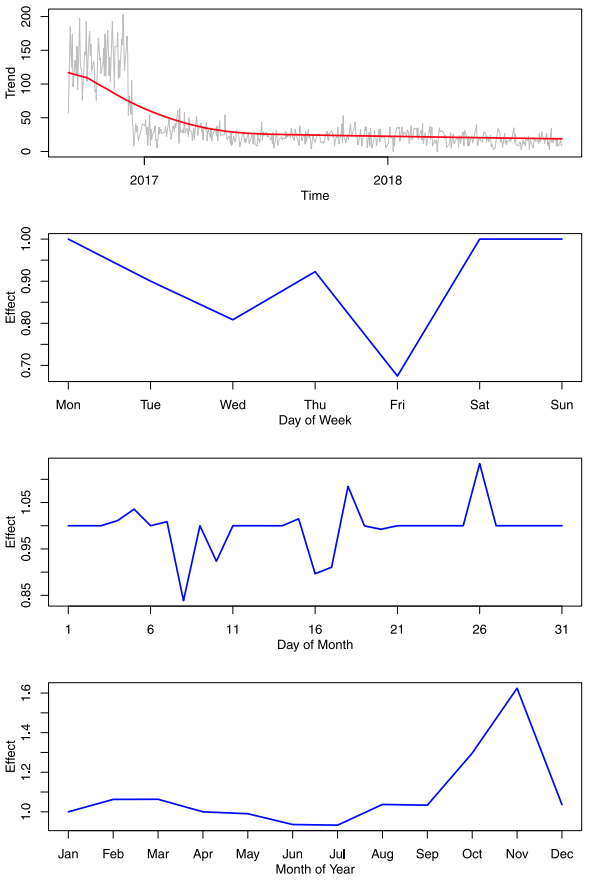


Fig. 5. Structural characteristics of time series B2. The time series is decomposed into trends and weekly, monthly, and yearly seasonal patterns.

2019) and *Stan* (Carpenter et al., 2017). *Stan* is known as a state-of-the-art platform for statistical modeling and high-performance statistical computation. In particular, *Stan* can be used to compute MAP estimates of model parameters and to perform full Bayesian inference via Markov chain Monte Carlo (MCMC) sampling. Hence, a few lines of *Stan* code, complemented by a *R* script to compute the matrices **X** and **Z**, suffice to express our Bayesian models.

For the remainder of this section, we use the following abbreviations to identify our models: When the target values are assumed to have a negative binomial distribution, our method is abbreviated by `NegBinom`, and when a normal distribution is assumed, our method is denoted as `Normal`.

5.3. Insights on the proposed models

In this subsection, we illustrate how well the proposed models fit the properties of the provided real-world data. The evaluation is based on representative datasets from the above-described restaurants A and B. In Section 5.3.1, the priors given by Eqs. (2)–(4) are assigned to the trend functions of the considered models. Further, the standard settings specified in (12) are assigned to the tuning parameters of the models. These are our recommendations for cases where model optimization has to take place automatically and at low computational cost. In Section 5.3.2, we illustrate the superior performance of more advanced approaches for a time series associated with restaurant B.

5.3.1. Standard approaches

At first, a representative time series A1 (compare to Section 5.1) corresponding to restaurant A is considered. In Fig. 6, the model fit is visualized for `Normal` (5) and in Fig. 7, the model fit of `NegBinom` (9) is shown. Detailed plots of the coefficients of the trend function and the seasonality function of both models are provided in the supplementary online resources. Compared to `Normal`, the seasonal function of `NegBinom` takes small values. The reason for this is that the seasonal function acts in a multiplicative way on the trend, $\mu_t = \exp(g(\delta(t)))\exp(s(t))$, in `NegBinom`, and in an additive way, $\mu_t = g(\delta(t)) + s(t)$, in `Normal`. For `Normal`, 94.63% of the observed sales lie within the 95% prediction intervals. The prediction intervals of `NegBinom` cover 94.97% of the sales. Thus,

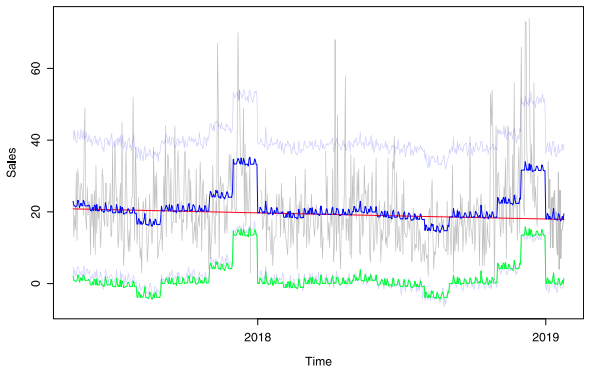


Fig. 6. Model fit of `Normal` on the time series A1. The trend function is plotted in red, the seasonal function is colored green, the expected value $\mathbb{E}(\mathbf{y}|\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ (see Eq. (13)) is shown in dark blue, and the 95% prediction intervals for the components of $\mathbf{y}$ are plotted in light blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

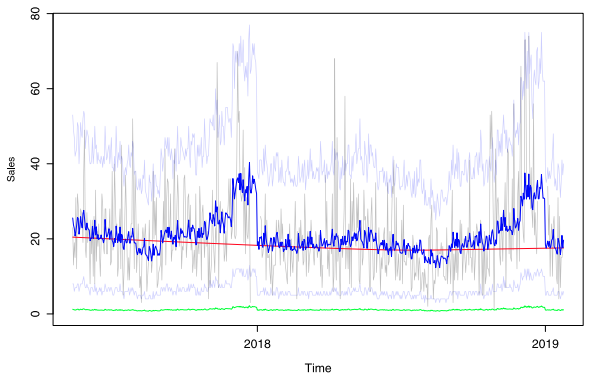


Fig. 7. Model fit of `NegBinom` on the time series A1. The trend function is plotted in red, the seasonal function is colored green, the expected value $\mathbb{E}(\mathbf{y}|\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ (see Eq. (13)) is shown in dark blue, and the 95% prediction intervals for the components of $\mathbf{y}$ are plotted in light blue. For the `NegBinom`, the exponential functions of the trend and the season are plotted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

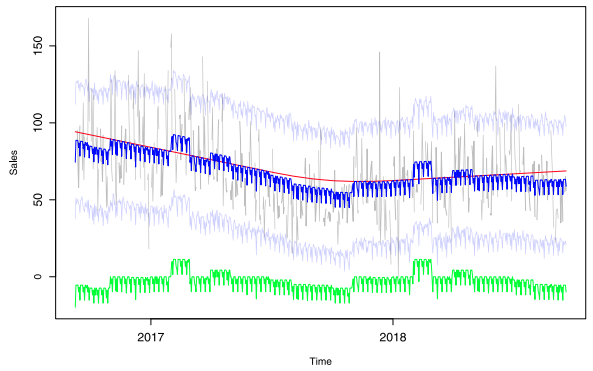


Fig. 8. Model fit of `Normal` on the time series B1. The trend function is plotted in red, the seasonal function is colored green, the expected value $\mathbb{E}(\mathbf{y}|\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ (see Eq. (13)) is shown in dark blue, and the 95% prediction intervals for the components of $\mathbf{y}$ are plotted in light blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

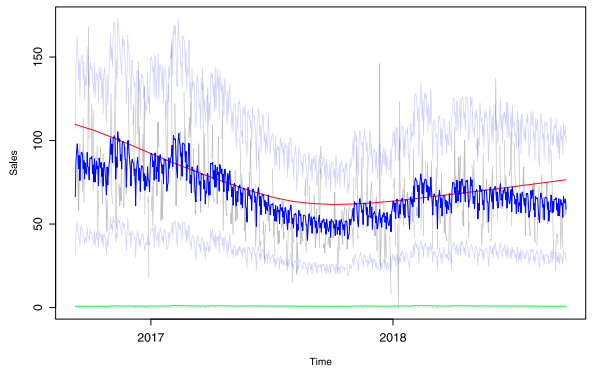


Fig. 9. Model fit of `NegBinom` on the time series B1. The trend function is plotted in red, the seasonal function is colored green, the expected value $\mathbb{E}(\mathbf{y}|\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ (see Eq. (13)) is shown in dark blue, and the 95% prediction intervals for the components of $\mathbf{y}$ are plotted in light blue. For the `NegBinom`, the exponential functions of the trend and the season are plotted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

both models provide a reasonable fit of the observed time series.

Now, a representative time series B1 (compare to Section 5.1) belonging to restaurant B is considered. In Figs. 8 and 9, the model fit is shown for `Normal` and for `NegBinom`. Detailed plots of the coefficients of the trend function and the seasonality function of both models are provided in the supplementary online resources. For `Normal`, 93.92157% of the observed sales lie within the corresponding 95% prediction intervals. The prediction intervals of `NegBinom` cover 95.29412% of the sales.

5.3.2. Advanced approaches

In this section, the time series B2 (compare to Section 5.1) corresponding to restaurant B is considered. This time series shows a drastic trend change within a short period of time. In Fig. 10, the model fit of `NegBinom` is visualized with different prior specifications of the trend function. We use the prior (3)–(4), that is inspired by the Bayesian Lasso, and also the horseshoe prior. In the case of the prior (3)–(4), the model is trained once with standard settings (12) and once according to the stepwise cross-validation (CV), as described in Section 4.5.3. Cross-validation is performed using six training and testing splits, where the testing sets always comprise 14 observations. In Figs. 10 and 11, we observe that the model encounters serious problems when the prior (3)–(4) is applied with the standard specification of the tuning parameters. The shrinkage of the trend coefficients is overly pronounced, such that the trend function fails to sufficiently model the abrupt decrease at the end of 2017. If cross-validation is performed, a better model fit is obtained. However, significant trend changes are detected nearly every month, which is an indication of overfitting. Due to the uniform shrinkage with the Lasso prior, insignificant trend changes cannot be shrunken sufficiently

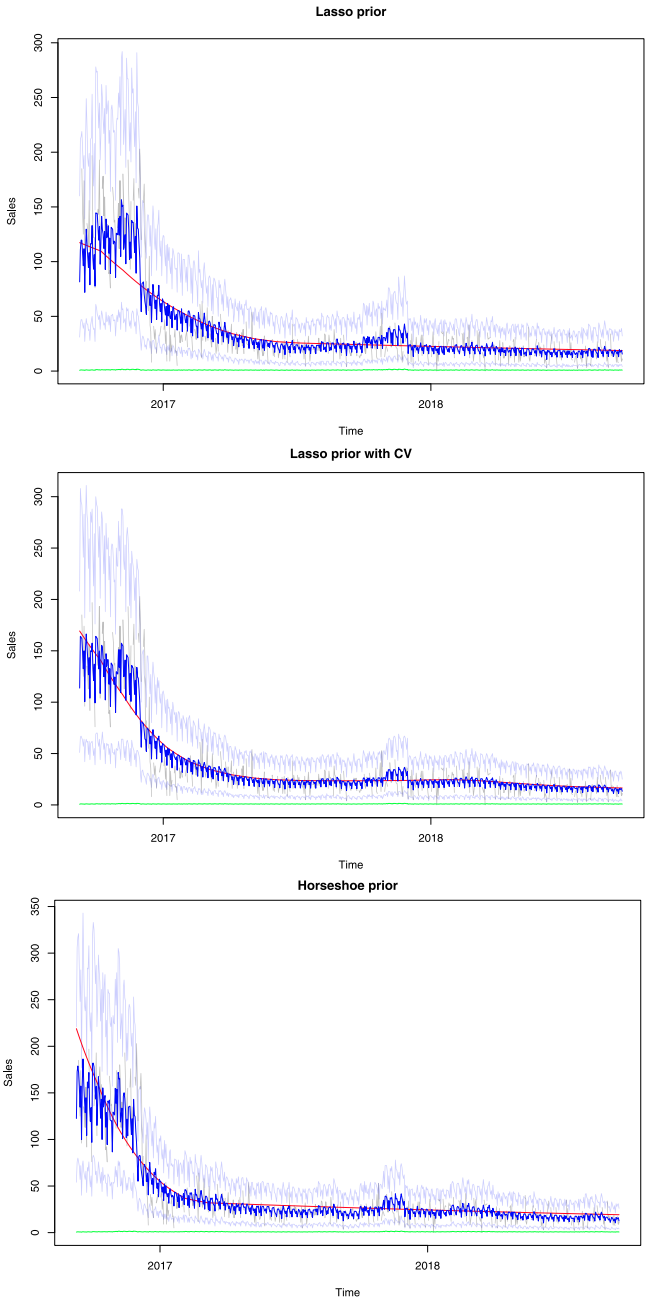


Fig. 10. Model fit of `NegBinom` on the time series B2. The trend function is plotted in red, the seasonal function is colored green, the expected value $\mathbb{E}(\mathbf{y}|\widehat{\boldsymbol{\beta}}, \widehat{\boldsymbol{\gamma}})$ (see Eq. (13)) is drawn in dark blue, and the 95% prediction intervals for the components of $\mathbf{y}$ are plotted in light blue. For the `NegBinom`, the exponential functions of the trend and the season are plotted. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to allow for the appearance of drastic changes. Finally, an application of the horseshoe prior results in a good model fit. The trend function includes few large coefficients in absolute values to model the abrupt trend change, while most of the coefficients are equal to zero.

5.4. Comparison against other prediction methods

In this section, the performance of our models is compared against other well-established time series forecasting approaches. The priors given by Eqs. (2)–(4) are

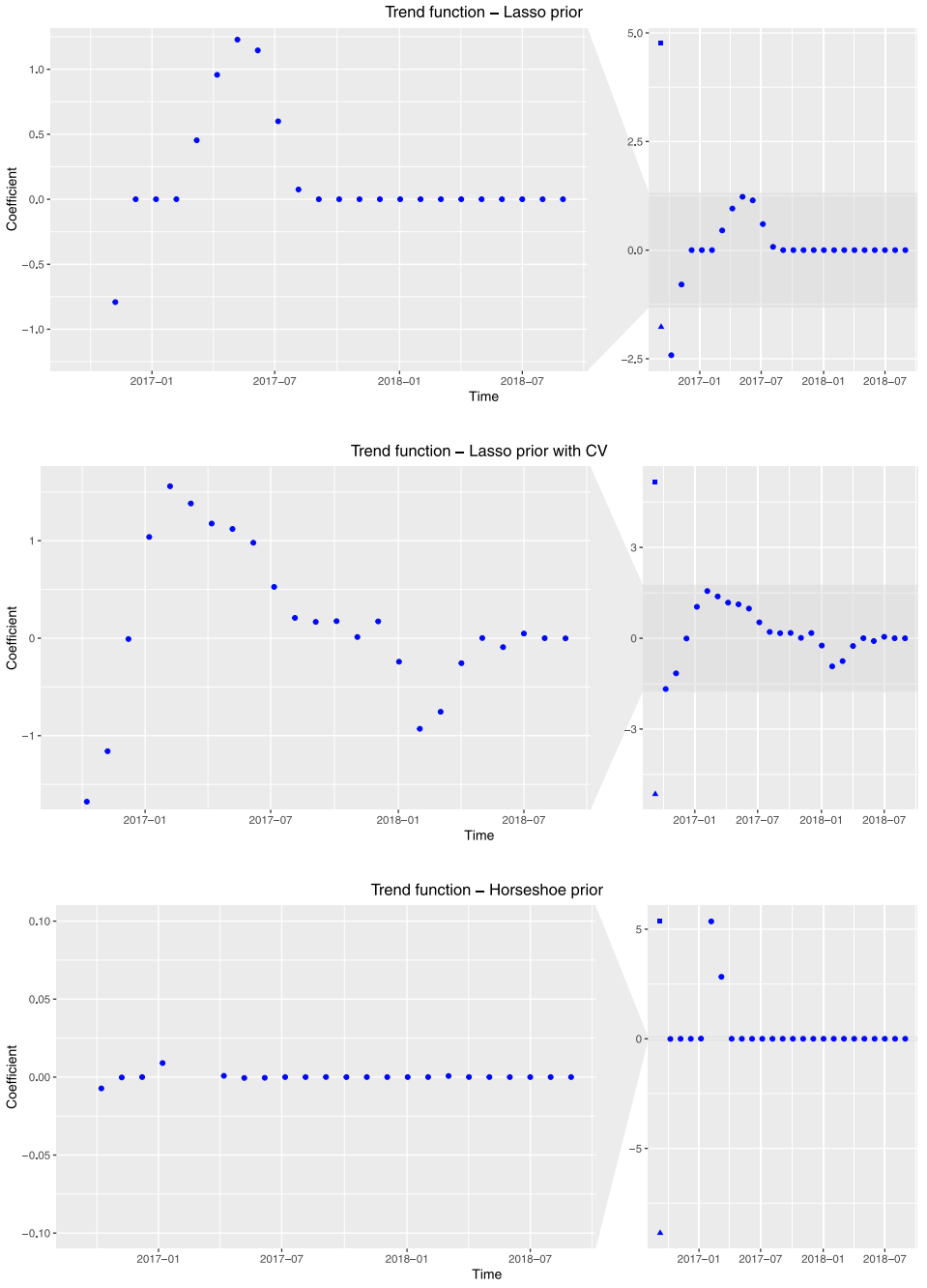


Fig. 11. Trend coefficients of `NegBinom` on the time series B2. Each coefficient is plotted against the time point at which it has a non-zero effect on the model for the first time. Since the coefficients of the global polynomial affect the model from the beginning, different symbols are used to visualize them (a square for the intercept γ_1 and a triangle for the coefficient γ_2).

assigned to the trend functions of our models and, further, the standard settings specified in (12) for the model tuning parameters are used.

For a given time series and a given accuracy measure, the performance of the considered model is evaluated using 15-fold cross-validation, as illustrated in Fig. 1. The size of the test datasets is specified as 14.

5.4.1. Forecast accuracy measures

Clearly, some measures are needed to compare the prediction quality of our approach against others. Let $O = \{t_1, \ldots, t_n\}$ denote the set of training time points, and let $F = \{t_1^*, \ldots, t_p^*\}$ denote the set of testing time points. Additionally, we denote the observed values by y_t and the predicted ones by $\widehat{y}_t$. A review of related work

(see Section 3) shows that the following are commonly used measures for point estimates (see, e.g., Xinliang and Dandan (2017) and Taylor and Letham (2018):

- Mean absolute deviation: $\text{MAD} = \frac{1}{p}\sum_{t\in F}|y_t - \widehat{y}_t|$,
- Mean squared error: $\text{MSE} = \frac{1}{p}\sum_{t\in F}(y_t - \widehat{y}_t)^2$,
- Mean absolute percentage error: $\text{MAPE} = \frac{1}{p}\sum_{t\in F}\left|\frac{y_t-\widehat{y}_t}{y_t}\right|$.

Another common measure is the mean forecast error:

$$\text{MFE} = \frac{1}{p}\sum_{t\in F}(y_t - \widehat{y}_t),$$

which is positive if a model tends to under-forecast, and negative otherwise. Except for MAPE, all of the measures above are scale dependent. That is, the larger the y_t the larger the error.

We compared our approach against others from the literature on several different time series, i.e., different menu items and restaurants, in order to obtain a broad comparison. Hence, it is desirable that the prediction quality measures are scale invariant. The above-mentioned scale-dependent measures are important because, in the restaurant industry, the calculation of the nominal and safety stocks are based on these measures. In the case of inferior estimates, there is a risk of inventory shortages or increased food waste. For this reason, scale-dependent measures are considered in Section 5.4.3, but with the note that the aggregated values presented in this section are dominated by those time series that show the largest values on average. The scale-invariant MAPE is not considered, since it cannot deal with zero values; i.e., y_t must be different from zero. Some more suitable measures are the mean arctangent absolute percentage error (MAAPE), proposed by Kim and Kim (2016), the weighted absolute percentage error (WAPE) (Chase, 1995), and the mean absolute scaled error (MASE) (Hyndman & Koehler, 2006; Makridakis et al., 2020):

- $\text{MAAPE} = \frac{1}{p}\sum_{t\in F}\arctan\left(\left|\frac{y_t-\widehat{y}_t}{y_t}\right|\right)$,
- $\text{WAPE} = \frac{\sum_{t\in F}|y_t-\widehat{y}_t|}{\sum_{t\in F}y_t}$,
- $\text{MASE} = \frac{\frac{1}{p}\sum_{t\in F}|y_t-\widehat{y}_t|}{\text{mean}(\text{abs}(\mathbf{y}-\mathbf{y}_m))}$,

with $m \in \mathbb{N}$, $\mathbf{y} = (y_{t_1+m}, \ldots, y_{t_n})^\mathsf{T}$ and $\mathbf{y}_m = (y_{(t_1+m)-m}, \ldots, y_{t_n-m})^\mathsf{T}$. MAAPE can be considered a modification of MAPE that allows for zero values, since it passes the absolute percentage errors to the arctan function. In contrast to MAPE, WAPE weights errors by the sales volume. MASE compares the MAD of the considered forecasting model (in the testing period F) with the MAD of the one-step seasonal naive method with season m (in the training period O). It should be mentioned that in Section 5.4.3, MASE is once used with $m = 1$ (MASE1) and once used with $m = 7$ (MASE7).

To evaluate the quality of the prediction intervals, the following measures are helpful: the prediction interval coverage probability (PICP) (Shen et al., 2018), the prediction interval normalized average width (PINAW) (Shen et al., 2018), and the mean scaled interval score (MSIS) (Gneiting & Raftery, 2007; Makridakis et al., 2020). It is desirable to have a small PINAW and a large PICP; see Khosravi et al. (2010). MSIS is designed to balance the relationship between PINAW and PICP. Note that, in this evaluation, 95% prediction intervals are always considered, since intervals with this confidence level are most commonly used in the business world; see Makridakis et al. (2020).

5.4.2. Prediction methods for comparison

Here, we briefly present the methods considered for the comparison against our proposed model. We adopted the selection taken by Taylor and Letham (2018).

- **Facebook's `prophet` method.** The `prophet` *R* package implements a modular regression model with interpretable parameters, see Taylor and Letham (2018). We discuss the model in more detail in Section 3. Within the performance comparison, the model is used once with automatic season detection (prophetA) and once with yearly and weekly seasonality (prophetS).
- **ARIMA.** The `auto.arima` function returns the best ARIMA model according to the AIC or BIC value. Here, we choose the Akaike information criterion (AIC).
- **Seasonal ARIMA.** The `auto.arima` function can also be used to determine the best seasonal ARIMA (SARIMA) model. To this aim, we specify the seasonality to be weekly, and we define the order of seasonal differencing as equal to 1. If the seasonal order is not specified, the consideration of a seasonality is decided by the `auto.arima` function, which might also return a non-seasonal model.
- **Exponential smoothing.** The `ets` function implements an exponential smoothing state space model. It fits a collection of exponential smoothing models and selects the best; see Hyndman et al. (2002). In this performance comparison, the seasonality of the exponential smoothing model is defined as weekly.
- **TBATS.** Trigonometric seasonality is based on Fourier series, Box–Cox transformation, ARMA errors, trend, and seasonal components; see Livera et al. (2011). In this comparison study, the TBATS model is specified to learn weekly, monthly, and yearly seasonalities.

Note that all above-mentioned forecasting models, except for `prophet`, are implemented in the `forecast` *R* package (Hyndman et al., 2019; Hyndman & Khandakar, 2008). Furthermore, ARIMA, SARIMA, ExpSmooth, and TBATS assume regularly spaced data points. Since the considered testing data (see Sections 2.1 and 5.1) contain missing data, the missing values are linearly interpolated and subsequently rounded before fitting one of these models.

5.4.3. Evaluation

Finally, we present how our approach performs compared to the ones described above. In Tables 3–5, averages of the performance measures described in Section 5.4.1 are given for restaurant A and restaurant B. For restaurant

Table 3
Performance comparison based on data from the category *main dishes* collected at restaurant A.

Measure	Normal	NegBinom	ProphetS	ProphetA	ARIMA	SARIMA	ExpSmooth	TBATS
MFE	0.255	0.209	0.158	0.336	0.110	0.120	0.058	0.065
MSE	14.133	13.974	14.001	15.680	15.538	16.893	14.925	15.282
MAD	2.681	2.675	2.693	2.800	2.823	2.964	2.755	2.810
WAPE	0.507	0.506	0.516	0.526	0.533	0.567	0.518	0.542
MAAPE	0.509	0.510	0.511	0.515	0.519	0.539	0.513	0.524
MASE1	0.779	0.777	0.789	0.811	0.818	0.866	0.798	0.829
MASE7	0.785	0.783	0.794	0.817	0.827	0.872	0.805	0.835
PICP	0.928	0.953	0.928	0.933	0.939	0.948	0.937	0.916
PINAW	1.211	1.168	1.243	1.308	1.368	1.551	1.350	1.252
MSIS1	5.638	4.761	5.666	5.868	5.811	5.787	5.572	6.236
MSIS7	5.695	4.798	5.706	5.927	5.877	5.845	5.630	6.261

Table 4
Performance comparison based on data from the category *desserts* collected at restaurant A.

Measure	Normal	NegBinom	ProphetS	ProphetA	ARIMA	SARIMA	ExpSmooth	TBATS
MFE	0.302	0.173	0.168	0.450	0.060	0.084	0.102	0.007
MSE	5.846	5.390	5.796	5.749	5.384	5.795	5.462	5.930
MAD	1.460	1.453	1.514	1.469	1.482	1.543	1.483	1.577
WAPE	0.863	0.863	0.882	0.844	0.881	0.904	0.862	0.957
MAAPE	0.766	0.771	0.789	0.785	0.783	0.810	0.782	0.806
MASE1	0.725	0.724	0.746	0.727	0.738	0.767	0.730	0.785
MASE7	0.737	0.735	0.757	0.739	0.750	0.779	0.742	0.799
PICP	0.953	0.968	0.950	0.956	0.964	0.968	0.965	0.957
PINAW	1.604	1.233	1.652	1.715	1.764	1.957	1.757	1.664
MSIS1	5.690	4.539	5.586	5.976	5.782	5.967	5.721	5.751
MSIS7	5.768	4.610	5.663	6.055	5.862	6.047	5.799	5.822

A, we only present the results for the categories *main dishes* (Table 3) and *desserts* (Table 4). The results for all other categories (see Table 1) can be found in the supplementary online resources.

For each single time series, the measures are computed based on 15-fold cross-validation. Considering the results for restaurant A, it can be seen that both of our models show a positive MFE for nearly all of the considered categories. This indicates that our approach tends to under-forecast the sales numbers. While also tending to under-forecast, the classical times series models (ARIMA, SARIMA, ExpSmooth, and TBATS) show the lowest and, thus, the best MFEs. However, none of the models under comparison show sufficiently large MFEs for us to consider their performance poor. In contrast to restaurant A, for restaurant B, the non-standard approaches (i.e., `Normal`, `NegBinom`, and `prophet`) obtain lower MFEs than the classical ones. In particular, our models show significantly lower values than all other methods under comparison.

Inspecting the MSEs and the MADs for restaurant A reveals that the proposed models often obtain the lowest errors. Whenever another model performs better than ours, the differences are rather small. For restaurant B, `NegBinom` obtains the lowest MAD and the second lowest MSE.

Switching the analysis from the scale-dependent metrics for point forecasts to the scale-invariant ones (WAPE, MAAPE, and MASE) shows that our approach, `NegBinom`, shows the best results overall. For restaurant B, `NegBinom` obtains significantly lower error values than all other considered approaches. This result is stronger than the result for the scale-dependent metrics, since the scale-invariant metrics are not dominated by the time series showing the highest number of sales.

Moreover, `NegBinom` always achieves the lowest mean scaled interval score. Therefore, the prediction intervals corresponding to this approach can be considered the most useful ones. The superior performance could be explained by a good coverage probability paired with tight interval widths.

Finally, it is important to connect the performance of the methods to the structural characteristics of the time series data. Section 5.1 shows that the individual time series have multiple seasonalities. Although weekly, monthly, and yearly seasonal effects are observed, the strength of each of these effects differs among menu items. In principle, a method capable of incorporating each of these seasonalities should return the best results. However, in the case of limited training data, one has to be careful. Allowing for too much flexibility, i.e., modeling each of the seasons, can lead to over-fitting. This may be the reason why TBATS often shows inferior results compared to some of the simpler forecasting approaches. On the other hand, considering only some of the seasonalities can also negatively impact model performance. For instance, the SARIMA model, which only considers weekly seasonalities, mostly returns worse point predictions than the simple ARIMA model. We assume that neglecting the remaining seasonalities makes it difficult to extract the

Table 5
Performance comparison based on data collected at restaurant B.

Measure	Normal	NegBinom	ProphetS	ProphetA	ARIMA	SARIMA	ExpSmooth	TBATS
MFE	−0.735	0.481	−2.864	−3.589	−5.061	−9.298	−5.191	−7.498
MSE	37 364.710	36 975.500	40 162.940	37 331.880	36 956.450	49 030.340	38 858.290	39 497.440
MAD	44.040	41.165	47.045	44.490	45.177	49.546	45.311	46.758
WAPE	0.903	0.736	0.954	0.816	0.810	0.937	0.838	0.855
MAAPE	0.614	0.580	0.641	0.602	0.615	0.588	0.621	0.621
MASE1	0.924	0.840	0.985	0.840	0.857	0.899	0.890	0.894
MASE7	0.938	0.847	0.999	0.851	0.871	0.916	0.900	0.905
PICP	0.913	0.910	0.894	0.943	0.941	0.939	0.948	0.914
PINAW	2.423	1.653	2.514	2.657	2.585	2.965	98.260	2.360
MSIS1	8.288	6.620	9.709	7.498	7.540	7.825	404.466	8.099
MSIS7	8.497	6.793	9.923	7.677	7.750	8.052	438.253	8.238

true weekly effects, especially if the amount of training data is limited. Thus, we conclude that the preferable approach is one which considers all of the mentioned seasonal effects, but at the same time, in terms of regularization, also penalizes model flexibility. This observation is consistent with the fact that our approach leads to the best predictions overall. `Normal` and `NegBinom` use shrinkage priors in order to automatically detect significant seasonalities and, further, to guarantee that their influence does not exceed a necessary level.

Finally, another reason for the superior performance is that all other methods somehow assume Gaussianity, whereas the negative binomial distribution is more appropriate to model count data. This might be the reason why `NegBinom` provides the most useful prediction intervals. It is likely that the use of the negative binomial distribution is a key factor. Similar findings are observed by Snyder et al. (2012), who also found that models based on the negative binomial distribution give the best prediction performance concerning a forecast of the intermittent demand for slow-moving inventories, viz., low-volume items such as auto parts. The negative binomial distribution is widely used for slow-moving inventory items, presumably because of its good performance in practice.

5.5. Evaluation of the day of month effect

While it is quite intuitive that the sales of menu items show weekly and yearly patterns, it is less evident that there are also effects that repeat monthly. As shown in Section 5.1 the proposed models detect monthly seasonalities within the time series data considered in this study. However, there is still the question of whether these effects significantly influence the model prediction accuracy. To answer this question, we repeated the evaluation performed in Section 5.4.3 for `Normal` and `NegBinom` with the day-of-the-month effect switched off. It turns out that the usage of this explanatory variable does not have a significant influence on the average model performance, whether for Restaurant A or Restaurant B. However, a more detailed analysis shows that there are some time series where consideration of the day of month leads to consistent improvements. For instance, time series A2 (see Section 5.1) is such a time series. In Table 6, the corresponding model accuracies are reported once with the day-of-the-month effect turned on and once with this effect turned off. We notice that the consideration of monthly patterns leads to superior accuracy. Since the additional model complexity introduced by the day-of-the-month effect has only a minor influence on the computation times, we conclude that its consideration is useful. However, we found that the model performance could only be improved in a fraction of the restaurant sales time series considered in this paper.

6. Conclusion & future work

In this work, a new approach for predicting future sales of menu items in restaurants and staff canteens was proposed. In particular, two Bayesian generalized additive models were presented. The first one assumes future sales to be normally distributed, while the second one uses the more appropriate negative binomial distribution. Both approaches use shrinkage priors to learn significant multiple seasonal effects and trend changes automatically. The features learned by the models have a straightforward human interpretation, which helps potential users to cultivate the necessary trust and confidence in the methodology. The performance of our approach was extensively evaluated and compared to other well-established forecasting methods. To do so, we analyzed two data sets retrieved from POS systems collected at a restaurant and a staff canteen. The evaluations showed that our approach often provided the best and most robust point predictions overall. Moreover, the prediction intervals of our model with the negative binomial distribution were notably more accurate than the ones obtained by the other methods.

Currently, our approach only takes into account POS data. This makes it universally applicable, but also limits the prediction quality that can be achieved. In future work, we plan to enrich the data source by including weather data, information regarding special events and holidays, and, expert knowledge from restaurant managers. Additionally, we want to extend our approach such that it can also predict on hourly basis. Accurate sales predictions on an hourly basis will allow for an optimization of workforce planning.

Table 6
Performance evaluation of the effect of the day of the month for time series A2.

Measure	Normal without monthly effect	Normal with monthly effect	NegBinom without monthly effect	NegBinom with monthly effect
MFE	−0.004	0.002	−0.011	−0.012
MSE	4.058	3.987	4.007	3.890
MAD	1.609	1.595	1.601	1.581
WAPE	0.596	0.591	0.594	0.586
MAAPE	0.593	0.590	0.592	0.588
MASE1	0.704	0.698	0.701	0.692
MASE7	0.742	0.735	0.738	0.728
PICP	0.957	0.950	0.969	0.969
PINAW	1.260	1.221	1.133	1.109
MSIS1	4.527	4.429	3.812	3.741
MSIS7	4.763	4.660	4.012	3.937

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2021.06.001.

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